# Gustavo Soares Pinto

MSc Candidate in Applied Physics · Quantum Computing & Machine Learning for Energy Systems

Universidade Federal da Integração Latino-Americana (UNILA) · Itaipu Parquetec · Foz do Iguaçu, Brazil

soaresgustavo.p@gmail.com · nerdsoares.github.io/Gustavo_Soares · orcid.org/0009-0009-6474-9233 · linkedin.com/in/soaresgustavop

## RESEARCH INTERESTS

Quantum computing for optimization and machine learning on near-term (NISQ) devices — QUBO and Ising formulations, quantum annealing, QAOA and variational quantum algorithms, and quantum machine learning — applied to the modelling, forecasting and operation of energy systems, including their deployment for real-time inference on edge hardware. Secondary interests in deep learning for multivariate industrial time series and in techno-economic assessment of hydrogen technologies.

## CURRENT PROJECT

### Mapping Green Hydrogen Infrastructure Planning onto NISQ Devices: A QAOA Study of Electrolyser Siting

Poster · School on Quantum Simulation in the NISQ era · ICTP-SAIFR / IFT-UNESP, São Paulo · Nov 2026

Where should electrolyser plants go in Brazil's Northeast? The siting problem is built entirely from open Brazilian data (ONS curtailment, OpenStreetMap substations, road distances, electricity prices), encoded as a 16-qubit QUBO and solved with QAOA in simulation and on IBM quantum hardware. The study maps which structural feature binds first: QAOA learns the constraints, but the cost differences between good plans are too small, inside the penalty-dominated energy scale, for the circuit to prefer the best one.

**16 qubits**
4 candidate sites × 3 demand hubs

**~140×**
valid plans found above chance (simulation)

**21× / 20× / 9×**
on IBM Heron (ibm_kingston), 1–3 layers

## EDUCATION

### Master of Science in Applied Physics

UNILA · 2025 – present · expected defense Mar–Apr 2027

- Dissertation (provisional): *Edge AI and Quantum Machine Learning for Real-Time Monitoring of Green Hydrogen Systems*. Advisor: Prof. Dr. Raphael Fortes Infante Gomes.
- Research focus: quantum machine learning applied to energy systems, with real-time inference on embedded hardware (Raspberry Pi).
- Full merit-based graduate fellowship from CAPES (Brazilian Ministry of Education), awarded through competitive national selection.

### Bachelor of Science in Engineering Physics

UNILA · 2019 – 2025

- Thesis: development of a pilot algorithm for predicting the hydrogen outlet temperature in the production plant at Itaipu Parquetec — a hybrid quantum–classical study benchmarking a quantum LSTM (QLSTM) against classical recurrent architectures on operational data from an alkaline electrolyzer. Advisor: Prof. Dr. Raphael Fortes Infante Gomes.

## RESEARCH EXPERIENCE

### Machine Learning Research Fellow

Itaipu Parquetec — Center for Hydrogen Technologies · 2024 – present

- Deep-learning models for predictive maintenance and operational forecasting of an alkaline water electrolyzer (LSTM and recurrent architectures in PyTorch, multivariate plant telemetry).
- Extended the forecasting pipeline to hybrid quantum–classical models, benchmarking quantum LSTM architectures against classical counterparts on the same dataset.
- Deep reinforcement learning for automated feature selection, with Optuna-based hyperparameter search, to forecast temperature, pressure and stack voltage.
- Designed and deployed a real-time predictive operations dashboard used by plant staff for decision support.
- Co-developed the techno-economic assessment of a prototype hydrogen-powered boat and contributed to its assembly; supported assembly and integration of a hydrogen power generator, with technical and economic viability models.

### Undergraduate Researcher — Self-Powered Systems

UNILA · 2021 – 2022

- Self-powered acquisition system for autonomous sensors; programmed pyranometers and pyrheliometers for solar-irradiance data acquisition and processing.

### Undergraduate Researcher — Astronautics

UNILA · 2020 – 2021

- Aerodynamic modelling of rocket components (nose cone and fins) in Autodesk Inventor, followed by physical prototyping.

## PUBLICATIONS

Journal articles

Pinto, G. S.; Silva, M. A.; Gomes, R. F. I.; et al. “Machine Learning Applied to Predictive Maintenance in Alkaline Electrolyzers.” *Matéria (Rio de Janeiro)*, 2026. **submitted**

Pinto, G. S.; Gomes, R. F. I. “Quantum and Classical Recurrent Models for Operational Forecasting in Alkaline Water Electrolysis.” 2026. **in preparation**

Pinto, G. S.; Gomes, R. F. I. “Machine Learning and Quantum Machine Learning for Predictive Maintenance in Hydrogen Production: A Systematic Review.” 2026. **in preparation**

Conference contributions

Pinto, G. S.; Tavares Junior, A. T.; Gomes, R. F. I. “Mapping Green Hydrogen Infrastructure Planning onto NISQ Devices: A QAOA Study of Electrolyser Siting.” School on Quantum Simulation in the NISQ era, ICTP-SAIFR / IFT-UNESP, São Paulo, Brazil, Nov 2026. (Poster)

Pinto, G. S.; Maciel, J. N.; Gomes, R. F. I. “Predictive Maintenance in Hydrogen Production: A Systematic Analysis with Traditional and Quantum Machine Learning.” 2nd Meeting of Applied Physics at the Triple Frontier, Foz do Iguaçu, Brazil, Nov 2025. (Poster)

Pinto, G. S.; Silva, M. A.; Gomes, R. F. I.; Teogenes, A. T.; Quispe, A. A.; Isernhagen, E. K.; Cantane, D. A.; Pereira, A. C. “Machine Learning Applied to Predictive Maintenance in Electrolyzers.” 4th Brazilian Hydrogen Congress, Brasília, Brazil, Oct 2025. (Poster)

## TEACHING & OUTREACH

### Outreach Instructor — Smart Solar Collector

UNILA · 2023 – 2024

- Designed and taught hands-on classes using Arduino and relay circuits, covering foundational physics and the operation of low-cost smart solar collectors.

### Outreach Project Member — Educational Materials

UNILA · 2022 – 2023

- Designed layouts and specified material properties for educational cards teaching the physical concepts of metallic materials.

## AWARDS & LEADERSHIP

### CAPES Graduate Fellowship

Brazilian Ministry of Education · 2025 – present

### Honorable Mention — V SIEPE

Week of Teaching, Research and Extension · Foz do Iguaçu · 2023

### Vice President — Academic Center of Engineering Physics (CAEF)

Apr 2024 – Apr 2025

## SKILLS & COURSEWORK

### Technical skills

- **Quantum computing:** Qiskit, PennyLane, D-Wave Ocean; QUBO/Ising formulations, quantum annealing, QAOA, variational circuits, QML (QLSTM, QSVM)
- **Deep learning:** PyTorch, TensorFlow, LSTM and recurrent architectures, deep reinforcement learning, Optuna
- **Data & tooling:** Python, LaTeX, scikit-learn, pandas, NumPy, Git
- **Engineering:** predictive maintenance, techno-economic assessment, hydrogen and electrolysis systems, microcontrollers, Arduino, Autodesk Inventor

### Graduate coursework

- Quantum Computing I & II (2025–2026)
- Quantum Mechanics
- Artificial Intelligence Applied to Scientific Research

### Languages

- Portuguese — native
- English — C1
- Spanish — C1
- Mandarin — A1

Last updated: November 2026 · References available upon request.